Variable Selection for Wasserstein Robust Hypothesis Testing

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Abstract

Variable selection for Wasserstein robust testing admits an exact finite formulation and a maximum-affine dual representation. Dual couplings generate global cuts.

1 Problem formulation

Throughout this paper, we consider the variable selection problem for robust hypothesis testing. Let $[D] = \{1, \ldots, D\}$ and d be a positive integer denoting the sparsity budget. Given a detector $f : \mathbb{R}^D \to \mathbb{R}$, we follow the convention in [2] to define the worst-case risk as

$$R(f) = \sup_{\mathbb{P}_0 \in \mathcal{P}_0, \mathbb{P}_1 \in \mathcal{P}_1} \left\{ \mathbb{E}_{P_0} \ell(-f(X)) + \mathbb{E}_{P_1} \ell(f(X)) \right\}, \tag{1}$$

where $\mathcal{P}_k, k = 0, 1$ denote the Wasserstein distance-based ambiguity sets, $\ell(\cdot)$ is the generating function that is nonnegative valued, non-decreasing, convex, and satisfies $\ell(0) = 1, \lim_{t \to -\infty} \ell(t) = 0$. For $k = 0, 1$, assume we have n available observations in the group k, which are denoted as $\{x_i^{(k)}\}_{i=1}^n$. We denote the corresponding empirical distribution as $\widehat{\mathbb{P}}_k$, and construct the Wasserstein ambiguity set as

$$\mathcal{P}_k = \left\{ \mathbb{P}_k : \mathcal{W}_1(\widehat{\mathbb{P}}_k, \mathbb{P}_k) \le \rho_k \right\}.$$

Here, $\mathcal{W}_1(\cdot, \cdot)$ refers to the 1-Wasserstein distance equipped with ℓ_1-norm.

For $S \subseteq [D]$, define the collection of detectors whose input only depends on coordinates belonging to S:

$$\mathcal{F}_S = \left\{ f : \mathbb{R}^D \to \mathbb{R}, f(x) = g(x[S]) \text{ for some measurable function } g : \mathbb{R}^{|S|} \to \mathbb{R} \right\}.$$

Note that for a given vector $x \in \mathbb{R}^D$, we use $x[i]$ to denote its i-th coordinate, and $x[S]$ to denote the contamination of its coordinates whose index belongs to S. Our proposed variable selection problem can be formulated as the following combinatorial optimization

$$\min_{S \subseteq [D], |S| \le d} \min_{f \in \mathcal{F}_S} R(f). \tag{2}$$

2 Finite reformulation and duality

2.1 The risk for a fixed set of coordinates

Fix a selection S. We first compute the best risk achievable using these coordinates; the outer minimization will then compare different selections. In the remainder, assume $\rho_0, \rho_1 > 0$ and that ℓ is finite on $\mathbb{R}$. All probability distributions have finite first moments. Pool the observations into a single list:

$$N = 2n, \qquad I_0 = [n], \qquad I_1 = \{n+1, \ldots, 2n\}, \qquad x_i = x_i^{(0)}, \quad x_{n+i} = x_i^{(1)} \quad (i \in [n]).$$

Thus every source observation has probability mass $1/n$ within its class. Write $\ell_0(s) = \ell(-s)$, $\ell_1(s) = \ell(s)$, and define

$$V(S) = \min_{f \in \mathcal{F}_S} R(f), \qquad \mathrm{OPT} = \min_{|S| \le d} V(S),$$

$$y_i = x_i[S], \qquad \Delta_{ir,j} = |x_i[j] - x_r[j]|, \qquad c_{ir}^S = \|y_i - y_r\|_1 = \sum_{j \in S} \Delta_{ir,j}.$$

Here $i, r \in [N]$ index observations and $j \in [D]$ indexes coordinates.

Restricting a detector to S is equivalent to testing on the projected observations y_i. This also preserves the ambiguity sets exactly. With $\widehat{Q}_k^S = n^{-1} \sum_{i \in I_k} \delta_{y_i}$, where δ_y is a unit mass at y, let

$$Q_k^S = \{Q : \mathcal{W}_1(Q, \widehat{Q}_k^S) \le \rho_k\}.$$

The distributions of $X[S]$ as $\mathbb{P}_k$ ranges over $\mathcal{P}_k$ are precisely Q_k^S. Projection cannot increase transport cost; conversely, any transport in the selected coordinates can be lifted to the full space by keeping the other coordinates of each source fixed. Consequently,

$$V(S) = \inf_{g \text{ measurable}} \sum_{k=0}^{1} \sup_{Q \in Q_k^S} \mathbb{E}_Q \ell_k(g(Y)).$$

Gao et al. [2, Proposition 1 and Theorem 3] establish minimax equality and an exact finite-support formulation for this convex-loss testing problem. Applied to the projected observations, their result reduces the optimized risk to transport between the pooled sample locations. For class k, let π_{ir} be the mass moved from y_i to y_r, and set

$$\mathcal{T}_k(S) = \left\{ \pi \ge 0 : \sum_{r=1}^{N} \pi_{ir} = \frac{1}{n} \ (i \in I_k), \quad \sum_{i \in I_k, r \in [N]} c_{ir}^S \pi_{ir} \le \rho_k \right\}.$$

If u_r is the detector score assigned to target y_r, then

$$V(S) = \inf_{u \in \mathbb{R}^N} \sum_{k=0}^{1} \max_{\pi \in \mathcal{T}_k(S)} \sum_{i \in I_k, r \in [N]} \pi_{ir} \ell_k(u_r). \tag{3}$$

The last sum is an expected loss: target r receives class-k mass $p_{kr} = \sum_{i \in I_k} \pi_{ir}$, so its contribution is $p_{kr} \ell_k(u_r)$. Exactness concerns the minimum over detectors; it does not assert that a worst-case distribution for every fixed detector is supported on the observations.

For fixed scores u, each maximization in (3) is a transportation linear program. Its dual uses a variable $a_i \in \mathbb{R}$ for the source-mass equality and a price $\lambda_k \ge 0$ for the transport budget. Define

$$J(a, \lambda) = \frac{1}{n} \sum_{i=1}^{N} a_i + \rho_0 \lambda_0 + \rho_1 \lambda_1.$$

Linear programming duality gives the finite convex problem

$$\begin{aligned} V(S) = \min_{a, u \in \mathbb{R}^N, \ \lambda \in \mathbb{R}_+^2} \quad & J(a, \lambda) \\ \text{subject to} \quad & \ell_k(u_r) \le a_i + \lambda_k c_{ir}^S, \quad i \in I_k, \ r \in [N], \ k = 0, 1. \end{aligned} \tag{4}$$

Each constraint bounds the loss at a possible target by a source term plus its transport cost. Thus (4) computes $\min_{f \in \mathcal{F}_S} R(f)$, and minimizing its value over $|S| \le d$ solves the original problem. Appendix A gives the projection argument, the precise finite minimax and LP-duality steps, and an optimal detector defined beyond the sample locations.

The later work of Xie et al. [3, Lemma 2 and Theorem 1] develops finite-support and extension results for randomized tests whose risk is the sum of the two error probabilities. Its transport objective is $\sum_r \min\{p_{0r}, p_{1r}\}$. For the general generating loss used here, the corresponding objective is $\sum_r \inf_s \{p_{0r} \ell(-s) + p_{1r} \ell(s)\}$, as in Gao et al. [2, Theorems 2–3]. This is the convex-loss result used to derive (4).

2.2 Representing the coordinate selection

Let $z_j = 1$ when coordinate j is selected and $z_j = 0$ otherwise. Following the logical-variable viewpoint of Bertsimas et al. [1], an unselected coordinate will force its associated continuous variables to zero. In our problem these variables are the two coordinate prices w_{0j}, w_{1j}, replacing the products $\lambda_0 z_j, \lambda_1 z_j$ in the loss constraints.

For $z \in [0,1]^D$, define

$$\begin{aligned} F(z) = \min_{a,u,\lambda,w} \quad & J(a,\lambda) \\ \text{subject to} \quad & \ell_k(u_r) \le a_i + \sum_{j=1}^{D} \Delta_{ir,j} w_{kj}, \quad i \in I_k,\ r \in [N],\ k = 0,1, \\ & 0 \le w_{kj} \le \lambda_k, \quad k = 0,1,\ j \in [D], \\ & \rho_0 w_{0j} + \rho_1 w_{1j} \le z_j J(a,\lambda), \quad j \in [D], \\ & a, u \in \mathbb{R}^N, \qquad \lambda \in \mathbb{R}^2_+. \end{aligned} \tag{5}$$

For fixed z this is a convex program. The last family of inequalities implements the selection: if $z_j = 0$, positivity of the radii forces $w_{0j} = w_{1j} = 0$. If $z_j = 1$, it adds no restriction, because the constraints with $r = i$ give $a_i \ge 0$ and hence

$$\rho_0 w_{0j} + \rho_1 w_{1j} \le \rho_0 \lambda_0 + \rho_1 \lambda_1 \le J(a,\lambda).$$

At $z = \mathbf{1}_S$, every feasible solution of (4) lifts to (5) by $w_{kj} = \lambda_k z_j$. Conversely, a feasible solution of (5) satisfies

$$\ell_k(u_r) \le a_i + \sum_{j \in S} \Delta_{ir,j} w_{kj} \le a_i + \lambda_k c^S_{ir}.$$

The objective is unchanged in both directions, so $F(\mathbf{1}_S) = V(S)$. This particular reformulation is derived for the testing problem here; the cited logical-constraint paper supplies the modelling and outer-approximation viewpoint.

2.3 A dual that is affine in the selection

The purpose of the next duality step is to obtain lower bounds that can be reused across selections. Introduce nonnegative multipliers Π_{ir} for the loss constraints and μ_j for the selection constraints in (5). The matrix Π is a scaled transport plan. To describe its objective and feasible set, write

$$m_{kr}(\Pi) = \sum_{i \in I_k} \Pi_{ir}, \qquad C_{kj}(\Pi) = \sum_{i \in I_k, r \in [N]} \Pi_{ir} \Delta_{ir,j},$$

$$H(\Pi) = \sum_{r=1}^{N} \inf_{s \in \mathbb{R}} \{ m_{0r}(\Pi) \ell(-s) + m_{1r}(\Pi) \ell(s) \}.$$

Here m_{kr} is the class-k mass at target r, C_{kj} is the transport expenditure in coordinate j, and each term of H is the infimum of the weighted loss at one target. Unlike $\pi \in \mathcal{T}_k(S)$, Π need not have total class mass one.

The Lagrangian calculation in Appendix B gives

$$\begin{aligned} F(z) = \max_{\Pi,\alpha,\mu} \quad & H(\Pi) \\ \text{subject to} \quad & (\Pi,\alpha,\mu) \in \mathcal{D}, \qquad \alpha + \mu^\top z = 1, \end{aligned} \tag{6}$$

where α scales the source masses and, with $(v)_+ = \max\{v, 0\}$,

$$\mathcal{D} = \left\{ (\Pi,\alpha,\mu) \ge 0 : \begin{array}{ll} \displaystyle\sum_{r=1}^{N} \Pi_{ir} = \alpha/n, & i \in [N], \\ \displaystyle\sum_{j=1}^{D} (C_{kj}(\Pi) - \rho_k \mu_j)_+ \le \alpha \rho_k, & k = 0,1 \end{array} \right\}. \tag{7}$$

The row equalities come from minimizing over a, while the positive-part inequalities come from minimizing over λ and w. These formulas are derived from (5), rather than imported from the references.

The normalization in (6) still depends on z. Since $\mathcal{D}$ is a cone, scaling a normalized tuple by $\tau \geq 0$ keeps it feasible and gives

$$H(\tau\Pi) = \tau H(\Pi), \qquad \max_{\tau \geq 0} \{2\sqrt{\tau H(\Pi)} - \tau\} = H(\Pi).$$

Optimizing the scale on each ray therefore yields

$$\boxed{F(z) = \max_{(\Pi,\alpha,\mu)\in\mathcal{D}} \{2\sqrt{H(\Pi)} - \alpha - \mu^\top z\}.} \tag{8}$$

Each feasible tuple now defines an affine lower bound for every selection. At a given z, a normalized optimizer $(\bar{\Pi}, \bar{\alpha}, \bar{\mu})$ must be multiplied by $F(z)$ to give an optimizer of (8). In particular, its affine slope is $-F(z)\bar{\mu}$, not $-\bar{\mu}$. Proposition B.1 proves the equivalence, attainment, and contact with F at the evaluation point.

3 Cutting-plane algorithm

The affine representation allows the outer-approximation strategy of Bertsimas et al. [1, Section 3.1]: solve the continuous problem for one selection, retain its affine lower bound, and use a binary linear program to choose the next selection. Let $\mathcal{Z}_d = \{z \in \{0,1\}^D : \mathbf{1}^\top z \leq d\}$. At z^t, take an optimizer (Π^t, α^t, μ^t) of (8). It defines

$$\begin{aligned} &c_t(z) = 2\sqrt{H(\Pi^t)} - \alpha^t - (\mu^t)^\top z, \\ &F(z) \geq c_t(z) = F(z^t) - (\mu^t)^\top (z - z^t), \qquad c_t(z^t) = F(z^t). \end{aligned} \tag{9}$$

Dual feasibility gives the lower bound, and dual optimality gives equality at z^t. Thus this is a supporting affine function, not merely an arbitrary lower bound; no differentiability of F is required. The next selection and the master lower bound solve

$$\begin{aligned} (z^{t+1}, \eta^{t+1}) \in \underset{z\in\mathcal{Z}_d,\ \eta\geq 0}{\arg\min} \quad & \eta \\ \text{subject to} \quad & \eta \geq F(z^s) - (\mu^s)^\top (z - z^s), \qquad s = 1, \ldots, t. \end{aligned} \tag{10}$$

With $U_t = \min_{s\leq t} F(z^s)$, all retained cuts lie below F, so

$$\eta^{t+1} \leq \mathrm{OPT} \leq U_t.$$

Stop when $U_t - \eta^{t+1} \leq \varepsilon$ and return the best evaluated selection. Appendix A recovers its detector from a solution of the finite convex problem.

Algorithm 1 Exact cutting-plane method for variable selection

Require: Problem data, budget d, tolerance $\varepsilon > 0$, and $z^1 \in \mathcal{Z}_d$.
Ensure: A subset S^*, its optimal detector f^*, and a risk interval of width at most ε.

$t \leftarrow 1, U \leftarrow +\infty$.
while true **do**
  Solve (8) at z^t and take an optimizer (Π^t, α^t, μ^t).
  if $F(z^t) < U$ **then**
    $(U, z^*) \leftarrow (F(z^t), z^t)$.
  end if
  Retain the cut (9) with index t.
  Solve (10) to obtain (z^{t+1}, η^{t+1}).
  if $U - \eta^{t+1} \leq \varepsilon$ **then**
    $S^* \leftarrow \{j : z^*_j = 1\}$.
    Solve (4) for S^* and construct g^* by (14).
    Set $f^*(x) = g^*(x[S^*])$ and **return** $(S^*, f^*, [\eta^{t+1}, U])$.
  end if
  $t \leftarrow t + 1$.
end while

A Equivalence with the finite testing problem

Proposition A.1 (Finite representation)**.** *For every $S \subseteq [D]$, the finite convex problem* (4) *attains the value $V(S) = \min_{f \in \mathcal{F}_S} R(f)$. Every optimal finite solution yields an optimal measurable detector on the full observation space.*

Proof. Let $\mathrm{Proj}_S\, x = x[S]$ and let $(\mathrm{Proj}_S)_\# \mathbb{P}$ denote the distribution of $X[S]$ under $\mathbb{P}$. Nonexpansiveness of Proj_S gives

$$\mathcal{W}_1((\mathrm{Proj}_S)_\# \mathbb{P}, \widehat{Q}_k^S) \le \mathcal{W}_1(\mathbb{P}, \widehat{\mathbb{P}}_k).$$

Conversely, disintegrate a coupling from $\widehat{Q}_k^S$ to $Q \in Q_k^S$ into source-indexed measures ν_i of mass $1/n$. For $i \in I_k$, define $L_i(y)[S] = y$ and $L_i(y)[S^c] = x_i[S^c]$. Then

$$\mathbb{P} = \sum_{i \in I_k} (L_i)_\# \nu_i, \qquad (\mathrm{Proj}_S)_\# \mathbb{P} = Q, \qquad \mathcal{W}_1(\mathbb{P}, \widehat{\mathbb{P}}_k) \le \sum_{i \in I_k} \int \|y - y_i\|_1 \, \nu_i(dy) \le \rho_k.$$

Repeated projected sources can be split into their original labels. Thus $(\mathrm{Proj}_S)_\# \mathcal{P}_k = Q_k^S$, and the projected robust testing problem has value $V(S)$.

For $S \neq \varnothing$, apply Proposition 1 and Theorem 3 of Gao et al. [2] on the projected Euclidean space. With $p_{kr} = \sum_{i \in I_k} \pi_{ir}^k$, their finite-support objective is the sum of the pointwise optimal weighted losses. It gives

$$\begin{aligned} V(S) &= \max_{\pi^0 \in \mathcal{T}_0(S),\, \pi^1 \in \mathcal{T}_1(S)} \sum_{r=1}^{N} \inf_{s \in \mathbb{R}} \{p_{0r} \ell(-s) + p_{1r} \ell(s)\} \\ &= \max_{\pi^0, \pi^1} \inf_{u \in \mathbb{R}^N} \sum_{k=0}^{1} \sum_{i \in I_k, r \in [N]} \pi_{ir}^k \ell_k(u_r) \\ &= \inf_{u \in \mathbb{R}^N} \sum_{k=0}^{1} \max_{\pi^k \in \mathcal{T}_k(S)} \sum_{i \in I_k, r \in [N]} \pi_{ir}^k \ell_k(u_r). \end{aligned} \tag{11}$$

In the second line the plans range over the same product set as in the first. The last equality is finite-dimensional minimax duality: the transport set is compact and convex, and the objective is continuous, affine in the plans, and convex in u. Coincident target locations can be merged; keeping their labels gives the same value. If $S = \varnothing$, the projected space is a singleton and both sides are $\min_s \{\ell(-s) + \ell(s)\} = 2$, so (3) still holds.

For fixed u and class k, the transport LP has Lagrangian

$$\begin{aligned} L_k &= \sum_{i \in I_k, r} \pi_{ir} \ell_k(u_r) + \sum_{i \in I_k} a_i \left(\frac{1}{n} - \sum_r \pi_{ir} \right) + \lambda_k \left(\rho_k - \sum_{i \in I_k, r} c_{ir}^S \pi_{ir} \right) \\ &= \frac{1}{n} \sum_{i \in I_k} a_i + \rho_k \lambda_k + \sum_{i \in I_k, r} \pi_{ir} [\ell_k(u_r) - a_i - \lambda_k c_{ir}^S]. \end{aligned} \tag{12}$$

Its supremum over $\pi \ge 0$ is finite exactly when all bracketed coefficients are nonpositive. LP strong duality therefore gives

$$\begin{aligned} \max_{\pi \in \mathcal{T}_k(S)} \sum_{i \in I_k, r} \pi_{ir} \ell_k(u_r) = \min_{(a_i)_{i \in I_k},\, \lambda_k \ge 0} \quad & \frac{1}{n} \sum_{i \in I_k} a_i + \rho_k \lambda_k \\ \text{subject to} \quad & \ell_k(u_r) \le a_i + \lambda_k c_{ir}^S, \quad i \in I_k,\ r \in [N]. \end{aligned} \tag{13}$$

Summing over the classes and minimizing over u proves (4). Its sublevel $J \le 2$ has $0 \le a_i \le 2n$ and $0 \le \lambda_k \le 2/\rho_k$. The constraints then bound both $\ell(u_r)$ and $\ell(-u_r)$. Since $\ell(t) \to \infty$ as $t \to \infty$, this sublevel is bounded and closed, and a minimum is attained.

To recover a detector, take a feasible (a, u, λ) in (4). At $y \in \mathbb{R}^{|S|}$, choose the smallest minimizing index in each class:

$$E_k(y) = \min_{i \in I_k} \{a_i + \lambda_k \|y_i - y\|_1\}, \qquad i_k = \min \arg\min_{i \in I_k} \{a_i + \lambda_k \|y_i - y\|_1\},$$
$$r_k = \|y_{i_k} - y\|_1, \qquad b = \|y_{i_0} - y_{i_1}\|_1 \le r_0 + r_1.$$

Define

$$g(y) = \begin{cases} (1-\gamma)u_{i_0} + \gamma u_{i_1}, & \gamma = \min\{1, r_0/b\}, \quad b > 0, \\ u_{i_0}, & b = 0. \end{cases} \tag{14}$$

If $b > 0$, then $\gamma b \le r_0$ and $(1-\gamma)b \le r_1$. Convexity and the four source–target constraints give

$$\ell_0(g(y)) \le (1-\gamma)\ell_0(u_{i_0}) + \gamma \ell_0(u_{i_1}) \le a_{i_0} + \lambda_0 \gamma b \le E_0(y),$$
$$\ell_1(g(y)) \le (1-\gamma)\ell_1(u_{i_0}) + \gamma \ell_1(u_{i_1}) \le a_{i_1} + \lambda_1 (1-\gamma) b \le E_1(y).$$

For $b = 0$, the same bounds follow from the constraints at target i_0. The finite tie rule makes g Borel measurable and $|g(y)| \le \max_r |u_r|$. Integrating its loss bounds against a transport coupling from $\widehat{Q}_k^S$ to any $Q \in \mathcal{Q}_k^S$ yields

$$\mathbb{E}_Q \ell_k(g(Y)) \le \frac{1}{n} \sum_{i \in I_k} a_i + \rho_k \lambda_k, \qquad R(f) \le J(a, \lambda), \quad f(x) = g(x[S]).$$

For an optimal finite solution, the right side is $V(S)$; the reverse inequality follows from $f \in \mathcal{F}_S$. Hence the recovered detector is optimal. □

B Dual representation and supporting affine functions

Proposition B.1 (Duality and subgradients)**.** (i) *For every $S \subseteq [D]$, $F(\mathbf{1}_S) = V(S)$. For every $z \in [0,1]^D$, both the primal minimum and the normalized dual maximum are attained, with*

$$0 < F(z) = \max_{\substack{(\Pi, \alpha, \mu) \in \mathcal{D} \\ \alpha + \mu^\top z = 1}} H(\Pi) \le 2. \tag{15}$$

(ii) *The value also has the attained representation*

$$F(z) = \max_{(\Pi, \alpha, \mu) \in \mathcal{D}} \{2\sqrt{H(\Pi)} - \alpha - \mu^\top z\}. \tag{16}$$

For a normalized optimizer $(\bar{\Pi}, \bar{\alpha}, \bar{\mu})$, an optimizer of this affine representation is

$$(\Pi, \alpha, \mu) = F(z)(\bar{\Pi}, \bar{\alpha}, \bar{\mu}), \qquad H(\Pi) = F(z)^2. \tag{17}$$

(iii) *The function F is convex on $[0,1]^D$. An affine optimizer (Π^t, α^t, μ^t) at z^t gives*

$$c_t(z) = 2\sqrt{H(\Pi^t)} - \alpha^t - (\mu^t)^\top z \le F(z),$$
$$c_t(z^t) = F(z^t), \qquad F(z) \ge F(z^t) - (\mu^t)^\top (z - z^t). \tag{18}$$

Thus $-\mu^t \in \partial F(z^t)$, where the subgradient inequality is understood on $[0,1]^D$.

Proof. (i) In either primal formulation, the constraints with $r = i$ imply $a_i \ge \ell_k(u_i) \ge 0$. If (a, u, λ) is feasible in (4), set $z = \mathbf{1}_S$ and $w_{kj} = \lambda_k z_j$. Then

$$\ell_k(u_r) \le a_i + \sum_j \Delta_{ir,j} w_{kj}, \qquad \sum_k \rho_k w_{kj} = z_j \sum_k \rho_k \lambda_k \le z_j J.$$

Conversely, a feasible solution of (5) with $z = \mathbf{1}_S$ satisfies

$$w_{kj} = 0 \ (j \notin S), \qquad \ell_k(u_r) \le a_i + \sum_{j \in S} \Delta_{ir,j} w_{kj} \le a_i + \lambda_k c_{ir}^S.$$

Both directions preserve J, proving $F(\mathbf{1}_S) = V(S)$.

To derive the dual, assign multipliers $\Pi_{ir}, \beta_{kj}, \mu_j \geq 0$ to the loss inequalities, the caps $w_{kj} \leq \lambda_k$, and the selection inequalities, respectively. Retain a, u free and $\lambda, w \geq 0$, and put $\alpha = 1 - \mu^\top z$. The Lagrangian is

$$\begin{aligned}\mathcal{L} &= J + \sum_{k, i \in I_k, r} \Pi_{ir} \left[\ell_k(u_r) - a_i - \sum_j \Delta_{ir,j} w_{kj} \right] + \sum_{k,j} \beta_{kj}(w_{kj} - \lambda_k) \\ &\quad + \sum_j \mu_j \left[\sum_k \rho_k w_{kj} - z_j J \right] \\ &= \sum_i a_i \left(\frac{\alpha}{n} - \sum_r \Pi_{ir} \right) + \sum_{r=1}^{N} \{ m_{0r}(\Pi) \ell(-u_r) + m_{1r}(\Pi) \ell(u_r) \} \\ &\quad + \sum_k \lambda_k \left(\alpha \rho_k - \sum_j \beta_{kj} \right) + \sum_{k,j} w_{kj} [\beta_{kj} + \rho_k \mu_j - C_{kj}(\Pi)].\end{aligned} \tag{19}$$

Its infimum over a, λ, w is finite precisely when

$$\sum_r \Pi_{ir} = \alpha/n, \qquad \sum_j \beta_{kj} \leq \alpha \rho_k, \qquad \beta_{kj} \geq (C_{kj}(\Pi) - \rho_k \mu_j)_+. \tag{20}$$

The row sums force $\alpha \geq 0$. Minimizing the remaining terms over u separates across targets and gives $H(\Pi)$. Choosing the smallest admissible β_{kj} in (20) gives exactly $\mathcal{D}$ and $\alpha + \mu^\top z = 1$, which proves the dual formula subject to strong duality.

For strong duality, first remove every variable w_{kj} with $z_j = 0$ and its selection row, since those variables are forced to zero. The reduced convex problem is strictly feasible at

$$u = 0, \qquad a_i = 2, \qquad \lambda_k = 1, \qquad w_{kj} = z_j/2 \quad (z_j > 0).$$

Indeed, the loss constraints and caps are strict and

$$\sum_k \rho_k w_{kj} = \frac{z_j}{2}(\rho_0 + \rho_1) < z_j(4 + \rho_0 + \rho_1) = z_j J \quad (z_j > 0).$$

Slater's theorem gives strong duality and dual attainment for the reduced problem. Restore a deleted coordinate using $\mu_j = \max_k C_{kj}(\Pi)/\rho_k$ and $\beta_{kj} = 0$; neither the objective nor normalization changes. Thus the full dual also attains $F(z)$, including at the boundary of the cube.

The constant feasible point $(a, u, \lambda, w) = (\mathbf{1}, 0, 0, 0)$ has value 2. The sublevel $J \leq 2$ bounds a, λ, w, and the loss constraints bound u on both sides, as in Appendix A. Hence a primal minimum exists. A zero value would force $a = \lambda = w = 0$ and therefore

$$2 = 2\ell(0) \leq \ell(-u_r) + \ell(u_r) \leq 0,$$

a contradiction. This proves (15).

(ii) The set $\mathcal{D}$ is a cone, and $H(\tau\Pi) = \tau H(\Pi) \geq 0$ for $\tau \geq 0$. For any feasible tuple let $b = \alpha + \mu^\top z$. If $b > 0$, normalization and part (i) imply $H(\Pi)/b \leq F(z)$, so

$$2\sqrt{H(\Pi)} - b = \frac{H(\Pi)}{b} - \left(\sqrt{\frac{H(\Pi)}{b}} - \sqrt{b} \right)^2 \leq F(z). \tag{21}$$

If $b = 0$, then $\alpha = 0$ and $\Pi = 0$, giving affine value zero. For a normalized optimizer, $H(\bar{\Pi}) = F(z)$ and $\bar{\alpha} + \bar{\mu}^\top z = 1$. Scaling it by $F(z)$ yields

$$H(F(z)\bar{\Pi}) = F(z)^2, \qquad 2\sqrt{H(F(z)\bar{\Pi})} - F(z)(\bar{\alpha} + \bar{\mu}^\top z) = F(z).$$

Thus the bound is attained, proving (16) and (17).

(iii) Since $\mathcal{D}$ is independent of z, (16) expresses F as a maximum of affine functions. It is therefore convex. A feasible tuple defines a lower bound for all z, and an optimal tuple at z^t attains equality there. Hence

$$c_t(z) = c_t(z^t) - (\mu^t)^\top (z - z^t) = F(z^t) - (\mu^t)^\top (z - z^t) \leq F(z). \qquad \square$$

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